YACHAY TECH UNIVERSITY

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# Mathematical Induction

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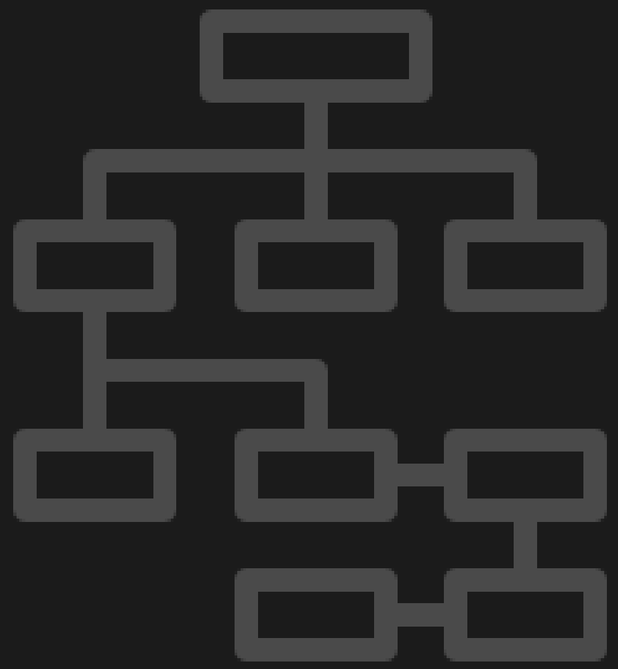
# Analysis of Algorithms

Sequence
Series
Summations
Mathematical induction



# Sequences

$\{2, 4, 6, 8, 10\}$

(ordered list)

- Even numbers greater or equal to 2 and smaller or equal to 10

Sequence is a function

$\{ x_n \}$ $\quad$ $x(n) = 2n$

| n | 1 | 2 | 3 | 4 | 5 | ... |
|---|---|---|---|---|---|---|
| x(n) | 2 | 4 | 6 | 8 | 10 | ... |

$x(n) = 2n \quad \forall n > 0$

$x(0) = 0$

# Series

A series is a summation of the elements of a list.

2+4+6+8+10  (*number)

$\sum$ "sigma notation"

$$\sum_{n=1}^{\infty} 2n = 2+4+6+8+\dots$$

## Summation Algebra

- $\sum_{i=1}^{n} 1 = \underbrace{1+1+\dots+1}_{n \text{ times}} = n$
- $\sum_{i=l}^{u} 1 = \underbrace{1+1+\dots+1}_{u-l+1 \text{ times}} = u-l+1$
- $\sum_{i=1}^{n} i = 1+2+3+\dots+n = \frac{n(n+1)}{2} \approx \frac{1}{2}n^2$
- $\sum_{i=1}^{n} i^2 = 1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6} \approx \frac{1}{3}n^3$

# Mathematical Induction

$$4 + 9 + 14 + 19 + \ldots + (5n-1) = (n/2)\,(3 + 5n)$$

Technique to prove that a property holds for all elements in a set. Typycally $\mathbb{N}$.

Two-step process:

1. Basis step
2. Induction step

# MI Ex (1): Series

① basis step
② induction step

$4 + 9 + 14 + 19 + \ldots + (5n-1) = (n/2)\,(3+5n)$

① basis step

$$4 \overset{?}{=} \frac{1}{2}(3+5\cdot 1)$$

$$4 \overset{\checkmark}{=} 4$$

② Induction step: Assume true for $n=k$, show true for $n=k+1$

Assume: $4+9+14+19+\ldots+(5k-1) \overset{\checkmark}{=} \frac{k}{2}(3+5k)$

Show: $4+9+14+19+\ldots+(5k-1)+[5(k+1)-1]$

$$\overset{?}{=} \frac{k+1}{2}\left[3+5(k+1)\right]$$

$$\frac{k}{2}(3+5k)+\left[5(k+1)-1\right] \overset{?}{=} \frac{k+1}{2}\left[3+5(k+1)\right]$$

$$\frac{3k}{2}+\frac{5}{2}k^2+5k+5-1 \overset{?}{=} \left(\frac{k+1}{2}\right)(5k+8)$$

$$5k^2+13k+4 = 5k^2+8k+5k+4$$

$$5k^2+13k+4 \overset{\checkmark}{=} 5k^2+13k+4 \quad \text{Q.E.D.}$$

# MI Ex (2): Series

$-1 + 2 + 5 + 8 + \ldots + (3n - 4) = (n/2)(3n - 5), \quad \forall\ \mathbb{N}$

- basis step

$n = 1$

$-1 = \frac{1}{2}(3 - 5)$

$-1 \overset{\checkmark}{=} -1$

- Induction step: Assume true for $k = n$, show true for $n = k + 1$

Assume: $-1 + 2 + 5 + 8 + \ldots + (3k - 4) \overset{\checkmark}{=} \frac{k}{2}(3k - 5)$

Show: $-1 + 2 + 5 + 8 + \ldots + (3k - 4) + [3(k+1) - 4] \overset{?}{=} \frac{k+1}{2}[3(k+1) - 5]$

$$\frac{k}{2}(3k - 5) + [3(k+1) - 4] \overset{?}{=} \frac{k+1}{2}[3(k+1) - 5]$$

$$\frac{3}{2}k^2 - \frac{5}{2}k + 3k + 3 - 4 \overset{?}{=} \frac{k+1}{2}(3k + 3 - 5)$$

$$\frac{3}{2}k^2 - \frac{1}{2}k - 1 \overset{?}{=} \overset{?}{=} \frac{k+1}{2}(3k - 2)$$

$$\overset{\checkmark}{=} \frac{3}{2}k^2 - \frac{1}{2}k - 1$$

# MI Ex (3): Sequence

$x(n) = 2x(n-1) + 3 \qquad \forall n >= 1$

$x(1) = 7$

Sol: $x(n) = 5(2n^n) - 3$

| x | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| x(n) | 7 | 17 | 37 | 77 | 157 |

$x(n) \overset{?}{=} 5(2^n) - 3$

(proposition)

• basis step

$n = 1$

$7 = 5(2^1) - 3$

$7 \overset{\checkmark}{=} 7$

• Induction step: Assume true for $n = k$, show true for $n = k+1$

Assume: $x(k) \overset{\checkmark}{=} 5 \cdot (2^k) - 3$

Show: $x(k+1) \overset{?}{=} 5 \cdot (2^{k+1}) - 3$

$2x(k+1-1) + 3 \overset{?}{=}$

$2x(k) + 3 \overset{?}{=}$

$2[5 \cdot (2^k) - 3] + 3 \overset{?}{=}$

$5 \cdot \underbrace{2 \cdot 2^k} - 6 + 3 \overset{?}{=}$

$5(2^{k+1}) - 3 \overset{\checkmark}{=} 5(2^{k+1}) - 3$

Q.E.D.

# MI Ex (4): Sequence

$x(n) = x(n - 1) + n \qquad \forall n > 0$

$x(0) = 0$

Sol: $x(n) = n(n + 1) / 2$

# What did we learn?

Sequences

Series

Summations

Mathematical induction

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# Solving Recurrences: Substitution

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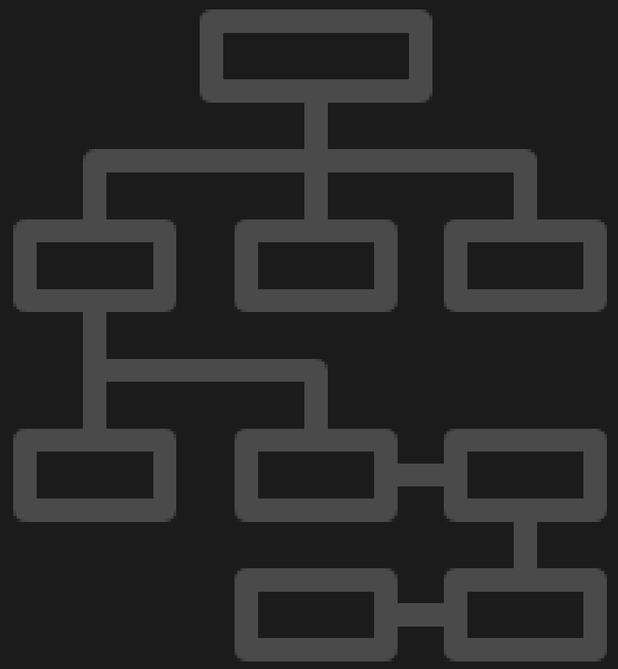
# Analysis of Algorithms

Forward substitution

Backward substitution



# Solving a recurrence

A recurrence relation is an equation that defines a sequence of values.
Future terms derive as a function of previous values.

$$\begin{cases} x(n) = x(n-1) + n & \leftarrow \text{recurrence} \\ x(0) = 0 & \leftarrow \text{I.C} \end{cases}$$

Solving a recurrence is

"Guess & confirm"

recurrence + I.C

$\downarrow$

explicit formula / closed formula

$\downarrow$

check (Mathematical Induction)

# Forward Substitution

$x(n) = 2\,x(n-1) + 1 \qquad \forall n > 1$

$x(1) = 1$

guess

$$\begin{cases} x(1) = 1 & = 1 \\ x(2) = 2\,x(2-1) + 1 = 2 \cdot 1 + 1 = & 3 \\ x(3) = 2\,x(3-1) + 1 = 2 \cdot 3 + 1 = & 7 \\ x(4) = 2\,x(4-1) + 1 = 2 \cdot 7 + 1 = & 15 \end{cases}$$

$\vdots$

$\boxed{x(n) = 2^n - 1} \quad \forall n = 1, 2, 3, \ldots$

explicit formula

M. I:

- basis step

$n = 1 \qquad 1 = 2^1 - 1$

$1 \overset{\checkmark}{=} 1$

- induction step: Assume true for $n = k$, show true $n = k+1$

Assume: $x(k) \overset{\checkmark}{=} 2^k - 1$

Show: $x(k+1) \overset{?}{=} 2^{k+1} - 1$

$x(k+1) = 2\,x(k+1-1) + 1$

$= 2\,x(k) + 1$

$2\,x(k) + 1 = 2^{k+1} - 1$

$2 \cdot (2^k - 1) + 1 =$

$2^{k+1} - 2 + 1 =$

$2^{k+1} - 1 \overset{\checkmark}{=} 2^{k+1} - 1$ Q.E.D

# Backward Substitution

$x(n) = x(n - 1) + n$

$x(0) = 0$

if $n = n - 1$

$x(n-1) = x(n-1-1) + n - 1$

$= x(n-2) + (n-1)$

$x(n) = x(n-2) + (n-1) + n$

if $n = n - 2$

$x(n-2) = x(n-2-1) + (n-2)$

$= x(n-3) + (n-2)$

$x(n) = x(n-3) + (n-2) + (n-1) + n$

$x(n) = x(n-1) + n$

$x(n) = x(n-2) + (n-1) + n$

$x(n) = x(n-3) + (n-2) + (n-1) + n$

general form

$x(n) = x(\cancel{n-i}^{0}) + (\cancel{n-i}^{0}+1) + (\cancel{n-i}^{0}+2) + \dots + n$

$n = 0$

$n - i = 0$

$n = i$

$x(n) = \cancel{x(0)}^{0} + 1 + 2 + 3 + 4 + \dots + n$

$= \frac{n(n+1)}{2}$ //

$x(n) = x(n-1) + n$

$x(0) = 0$

Sol: $x(n) = \frac{n(n+1)}{2}$

• basis step

$n = 0$

$x(0) = \frac{0(0+1)}{2}$

$0 \overset{\checkmark}{=} 0$

• induction step

Assume true $n=k$, show true for $n=k+1$

Assume: $x(k) \overset{\checkmark}{=} \frac{k(k+1)}{2}$

Show: $x(k+1) \overset{?}{=} \frac{(k+1)(k+1+1)}{2}$

$\overset{?}{=} \frac{(k+1)(k+2)}{2}$

$x(k+1-1) + k+1 =$

$x(k) + (k+1) =$

$\frac{k(k+1)}{2} + (k+1) =$

$\frac{k(k+1) + 2(k+1)}{2} =$

$\frac{(k+1)(k+2)}{2} = \frac{(k+1)(k+2)}{2}$ Q.E.D.

# Q & A

**FOR QUESTIONS OR COMMENTS**

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